"Ultrafast Laser-driven dynamics in 2D material nanoribbons"

This project aims to develop theory of laser-driven ultrafast electron dynamics in 2D materials nanoribbons. These are 1D-periodic structures with edge states that can have topological properties and form conducting channels with enhanced transport. Dynamics of exciting and driving currents through these states is largely unexplored on femtosecond timescales (we recently explored this in 2D flakes - https://arxiv.org/abs/2604.21121, in press in ACS Photonics). Students will develop theoretical tight-binding models that describe these systems, including the surface and spin-orbit coupling, and use them to explore time-dependent dynamics in the electronic states driven by the laser. We aim to understand how to excite these states effectively, and enhance their robustness to dissipation and scattering.

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